

TUTORIAL 1

1. With an aid of a diagram, discuss in details the architecture life cycle/activities.
2. What is software architecture, and how does it differ from other forms of design (detailed design and construction design)?
3. Software architecture describes a high-level design view of a software system. What are the advantages of explicitly describing the architecture independently from the implementation?
4. Discuss how software architecture serves as a basis for analysis. What about decision-making? What kinds of decision-making does software architecture empower?
5. Discuss the differences between *System architecture* and *Enterprise architecture*.
6. Discuss the importance of Software Architecture?
7. Explain architectural design, high level design and detailed design.
8. A newly established software development house has received its first business that required the system to develop an online toy selling system. The director requests the development team to proceed to coding in order to deliver the system to customer as earlier as possible. However, the software manager insists to produce an initial architecture of the system before start the coding. The software manager also insists to have a presentation of the system's structure in different level of details.
 - (a) Explain why the software manager insist to have an initial architecture of the system? Provide **THREE (3)** answers. Your answers should base on the software engineering perspective.
 - (b) Explain why the software manager requires different level of details on the architecture?
9. A bus ticketing system is proposed to the ABC bus company for the following objectives:
 - To increase the sales of bus tickets through online reservation.
 - To provide more convenient way to customers for ticket reservation.
 - To centralize the information regarding bus ticketing of all branches.

- (a) During the requirement engineering phase, the requirements from the users or stakeholders is confirmed and the software architect found that the most of the requirements from the users/stakeholders are not really help in shaping the architecture and even not useful in determine the requirement of architecture. Discuss Two (2) ways of why this divergence (mismatch) manifests.
- (b) Why business goals and objectives play an important role in gathering architectural requirement? Justify your answer.